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Lecture 02A
Before-Tutorial-Session
Tutorial 1

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Vector Space. $(\mathcal{V}, \oplus, \otimes)$ is a vector space if all the following hold.

- **(Closed under $\oplus$).** For all $\boldsymbol{u}, \boldsymbol{v} \in \mathcal{V}$, we have $\boldsymbol{u} \oplus \boldsymbol{v} \in \boxed{?}$ $\mathcal{V}$
- **(Closed under $\otimes$).** For all $\lambda \in \mathbb{F}, \boldsymbol{u} \in \mathcal{V}$, we have $\lambda \otimes \boldsymbol{u} \in \boxed{?}$ $\mathcal{V}$

Vector Addition $\oplus$

- ($\oplus$ **is commutative**). For all $\boldsymbol{u}, \boldsymbol{v} \in \boxed{?}$
$$\boldsymbol{u} \oplus \boldsymbol{v} = \boldsymbol{v} \oplus \boldsymbol{u}.$$
- ($\oplus$ **is associative**). For all $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w} \in \boxed{?}$
$$(\boldsymbol{u} \oplus \boldsymbol{v}) \oplus \boldsymbol{w} = \boldsymbol{u} \oplus (\boldsymbol{v} \oplus \boldsymbol{w}).$$
- **(Additive Identity).** There exists $\mathbf{0} \in \mathcal{V}$ such that
$$\boldsymbol{u} \oplus \mathbf{0} = \mathbf{0} \oplus \boldsymbol{u} = \boldsymbol{u} \text{ for all } \boldsymbol{u} \in \boxed{?}$$
- **(Additive Inverse).** For all $\boldsymbol{u} \in \boxed{?}$ there exists $\boldsymbol{u}^* \in \boxed{?}$ such that
$$\boldsymbol{u} \oplus \boldsymbol{u}^* = \boldsymbol{u}^* \oplus \boldsymbol{u} = \mathbf{0}.$$

Scalar Multiplication $\otimes$

- ($\otimes$ **is associative**). For all $\lambda, \mu \in \mathbb{F}, \boldsymbol{u} \in \boxed{?}$,
$$\lambda \otimes (\mu \otimes \boldsymbol{u}) = (\lambda\mu) \otimes \boldsymbol{u}.$$
- **(Multiplicative Identity).** Consider $1 \in \mathbb{F}$. Then
$$1 \otimes \boldsymbol{u} = \boldsymbol{u} \text{ for all } \boldsymbol{u} \in \boxed{?}.$$
- **(Distributive-1).** For all $\lambda \in \mathbb{F}, \boldsymbol{u}, \boldsymbol{v} \in \boxed{?}$
$$\lambda \otimes (\boldsymbol{u} \oplus \boldsymbol{v}) = (\lambda \otimes \boldsymbol{u}) \oplus (\lambda \otimes \boldsymbol{v}).$$
- **(Distributive-2).** For all $\lambda, \mu \in \mathbb{F}, \boldsymbol{u} \in \boxed{?}$
$$(\lambda + \mu) \otimes \boldsymbol{u} = (\lambda \otimes \boldsymbol{u}) \oplus (\mu \otimes \boldsymbol{u}).$$

Fill in the blank. All is $\mathcal{V}$

Question 1. Consider $\mathcal{V} = \mathbb{R}^* = \{\boldsymbol{r} \in \mathbb{R} : \boldsymbol{r} \geq 0\}$ with vector addition $\boldsymbol{a} \oplus \boldsymbol{b} = \sqrt{\boldsymbol{ab}}$ for all $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^*$, and scalar multiplication $\lambda \otimes \boldsymbol{a} = \lambda^2 \boldsymbol{a}$ for $\boldsymbol{a} \in \mathbb{R}^*, \lambda \in \mathbb{R}$.
In this question, check which axiom holds for the above operations $\oplus$ and $\otimes$. If an axiom does not hold, give an example to illustrate it.

- Is $\oplus$ **is commutative?**

Need to show all cases!

Is the Albert's solution correct? No.

Albert's Solution: Yes. Set $\boldsymbol{a} = 4$ and $\boldsymbol{b} = 1048576$ and we want to show

$$\boldsymbol{a} \oplus \boldsymbol{b} = \boldsymbol{b} \oplus \boldsymbol{a}.$$

- **LHS** is $4 \oplus 1048576 = \sqrt{4(1048576)} = 2048.$
- **RHS** is $1048576 \oplus 4 = \sqrt{(1048576)4} = 2048.$

Since LHS=RHS, we showed $\oplus$ is commutative.

To modify:
- need to forget the fact that a and b are set values.

Proof: → Correct one!

For $\underset{\sim}{a}, \underset{\sim}{b} \in V$,

$LHS = \underset{\sim}{a} \oplus \underset{\sim}{b} = \sqrt{\underset{\sim}{a} \cdot \underset{\sim}{b}}$

$RHS = \underset{\sim}{b} \oplus \underset{\sim}{a} = \sqrt{\underset{\sim}{b} \cdot \underset{\sim}{a}}$

Since $\underset{\sim}{a} \cdot \underset{\sim}{b} = \underset{\sim}{b} \cdot \underset{\sim}{a}$ as numbers,

So, $\sqrt{\underset{\sim}{a} \cdot \underset{\sim}{b}} = \sqrt{\underset{\sim}{b} \cdot \underset{\sim}{a}} \Rightarrow LHS = RHS$

$\therefore \underset{\sim}{a} \oplus \underset{\sim}{b} = \underset{\sim}{b} \oplus \underset{\sim}{a}$

Question 1. Consider $\mathcal{V} = \mathbb{R}^* = \{\boldsymbol{r} \in \mathbb{R} : \boldsymbol{r} \geq 0\}$ with vector addition $\boldsymbol{a} \oplus \boldsymbol{b} = \sqrt{\boldsymbol{ab}}$ for all $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^*$, and scalar multiplication $\lambda \otimes \boldsymbol{a} = \lambda^2 \boldsymbol{a}$ for $\boldsymbol{a} \in \mathbb{R}^*$, $\lambda \in \mathbb{R}$.
In this question, check which axiom holds for the above operations $\oplus$ and $\otimes$. If an axiom does not hold, give an example to illustrate it.

- Is $\oplus$ **is associative**?

Is the Albert's solution correct?

Sol$^{\underline{n}}$ is correct

Albert's Solution: No. Set $\boldsymbol{a} = 1$, $\boldsymbol{b} = 4$ and $\boldsymbol{c} = 9$, we want to show

$$(\boldsymbol{a} \oplus \boldsymbol{b}) \oplus \boldsymbol{c} \neq \boldsymbol{a} \oplus (\boldsymbol{b} \oplus \boldsymbol{c}).$$

- **LHS** is $(1 \oplus 4) \oplus 9 = 2 \oplus 9 = \sqrt{18}$.
- **RHS** is $1 \oplus (4 \oplus 9) = 1 \oplus 6 = \sqrt{6}$.

Since LHS is not RHS, we showed $\oplus$ is NOT associative.

→ This is a counterexample

*usually if a statement is wrong, you throw in any values, highly likely can show it's wrong.

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Can you do Question 1 from Tutorial 1? Both ANSWERS are correct.

1. **Yes, I can!** In fact, I have already done it!
2. **Yes, I can!** I haven't already done it. But I will do it!

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Theorem. **(Conditions for Subspace)** Let $\mathcal{V}$ be a vector space.

A subset $\mathcal{U}$ of $\mathcal{V}$ is a subspace if and only if $\mathcal{U}$ satisfies the following conditions.

- **(Zero Vector)**. $\mathbf{0}$ belongs to ? $\mathcal{U}$
- **(Closure under Vector Addition)**. For all $\boldsymbol{u}, \boldsymbol{v} \in \mathcal{U}$, we have $\boldsymbol{u} \oplus \boldsymbol{v} \in$? $\mathcal{U}$
- **(Closure under Scalar Multiplication)**. For $\lambda \in \mathbb{F}$, $\boldsymbol{u} \in \mathcal{U}$, we have $\lambda \otimes \boldsymbol{u} \in$? $\mathcal{U}$

Fill in the blank. Know that: $\mathcal{V}$ is a vector space

WTS: $\mathcal{U}$ is also a vector space

Examples of Subspaces

$\to$ we do this today

Proposition. $\mathcal{S}_{n\times n}(\mathbb{F})$ is a subspace of $(\mathcal{M}_{n\times n}(\mathbb{F}), \oplus, \otimes)$.

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Proposition. $\mathcal{P}_d(\mathbb{F})$ is a subspace of $(\mathcal{F}(\mathbb{F}, \mathbb{F}), \oplus, \otimes)$.

Proposition. The set of solutions of a homogeneous system is a subspace of $(\mathbb{F}^n, \oplus, \otimes)$.

Proposition. $\mathcal{S}_{n\times n}$ is a subspace of $\mathcal{M}_{n\times n}$.

$S_{n\times n}(\mathbb{F}) \triangleq \{$ symmetric $n\times n$ matrices whose entries belong to $\mathbb{F}\}$

$\rightarrow$ A matrix is symmetric if it is of the form $\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{12} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{nn} \end{bmatrix}$

$\rightarrow$ A matrix is symmetric iff $\hat{A} = \hat{A}^T$

Proposition. $\mathcal{S}_{n\times n}$ is a subspace of $\mathcal{M}_{n\times n}$.

$\boldsymbol{A}$ is symmetric iff $\boldsymbol{A} = \boldsymbol{A}^T$.

Theorem. (Conditions for Subspace)

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- **(Closure under Vector Addition).**

For all $\boldsymbol{u}, \boldsymbol{v} \in \mathcal{U}$, we have $\boldsymbol{u} \oplus \boldsymbol{v} \in \mathcal{U}$.

- **(Closure under Scalar Multiplication).**

For $\lambda \in \mathbb{F}$, $\boldsymbol{u} \in \mathcal{U}$, we have $\lambda \otimes \boldsymbol{u} \in \mathcal{U}$.

· What is $\underset{\sim}{0}$?

→ a zero matrix of dimension $n \times n$.

$\hat{0} = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & \cdots & 0 \end{bmatrix}$ is symmetric

$\therefore$ $\underset{\sim}{0}$ belongs to $\mathcal{U}$.

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- **(Closure under Scalar Multiplication).**

For $\lambda \in \mathbb{F}$, $\boldsymbol{u} \in \mathcal{U}$, we have $\lambda \otimes \boldsymbol{u} \in \mathcal{U}$.

Suppose $\hat{A}, \hat{B} \in \mathcal{S}_{n\times n}$

WTS $\hat{A} \oplus \hat{B} \in \mathcal{S}_{n\times n}$

What should we show?

$\rightarrow$ $\hat{A} \oplus \hat{B}$ is a symmetric matrix

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For $\lambda \in \mathbb{F}$, $\boldsymbol{u} \in \mathcal{U}$, we have $\lambda \otimes \boldsymbol{u} \in \mathcal{U}$.

WTS $\hat{A} \oplus \hat{B}$ is symmetric.

Suppose $\hat{A}$ and $\hat{B}$ are symmetric

$\Rightarrow \hat{A} = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \\ a_{1n} & \cdots & a_{nn} \end{bmatrix}$ & $\hat{B} = \begin{bmatrix} b_{11} & \cdots & b_{1n} \\ \vdots & & \\ b_{1n} & \cdots & b_{nn} \end{bmatrix}$

$\Rightarrow \hat{A} \oplus \hat{B} = \begin{bmatrix} a_{11}+b_{11} & \cdots & a_{1n}+b_{1n} \\ \vdots & \ddots & \vdots \\ a_{1n}+b_{1n} & \cdots & a_{nn}+b_{nn} \end{bmatrix}$

$\therefore \hat{A} \oplus \hat{B}$ is symmetric

So, $\mathcal{S}$ is closed under $\oplus$

Proposition. $\mathcal{S}_{n\times n}$ is a subspace of $\mathcal{M}_{n\times n}$.

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- **(Closure under Scalar Multiplication).**

For $\lambda \in \mathbb{F}$, $\boldsymbol{u} \in \mathcal{U}$, we have $\lambda \otimes \boldsymbol{u} \in \mathcal{U}$.

Suppose $\lambda \in \mathbb{F}$ & $\hat{A} \in \mathcal{S}_{n\times n}$

What to show?

$\rightarrow$ $\lambda \otimes \hat{A}$ is a symmetric matrix.

Proposition. $\mathcal{S}_{n\times n}$ is a subspace of $\mathcal{M}_{n\times n}$.

$\boldsymbol{A}$ is symmetric iff $\boldsymbol{A} = \boldsymbol{A}^T$.

WTS: $\lambda \otimes \hat{A}$ is symmetric

Theorem. **(Conditions for Subspace)**

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- **(Closure under Scalar Multiplication).**

For $\lambda \in \mathbb{F}$, $\boldsymbol{u} \in \mathcal{U}$, we have $\lambda \otimes \boldsymbol{u} \in \mathcal{U}$.

Suppose $\hat{A}$ is symmetrical & let $\lambda \in \mathbb{F}$

$$\hat{A} = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{1n} & \cdots & a_{nn} \end{bmatrix}$$

$$\lambda \otimes \hat{A} = \begin{bmatrix} \lambda a_{11} & \cdots & \lambda a_{1n} \\ \vdots & & \\ \lambda a_{1n} & \cdots & \lambda a_{nn} \end{bmatrix}$$

$\therefore \lambda \hat{A}$ is symmetric

$\therefore$ All 3 conditions met.

$\therefore$ S is a subspace

What is $(\boldsymbol{A} + \boldsymbol{B})^T$?

$$(A+B)^T = A^T + B^T$$

What is $(\lambda \boldsymbol{A})^T$?

$$(\lambda A)^T = \lambda A^T$$

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$\Rightarrow$ got 10 ciao

A is symmetric iff $A = A^T$.

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- **(Closure under Scalar Multiplication).**

For $\lambda \in \mathbb{F}$, $\boldsymbol{u} \in \mathcal{U}$, we have $\lambda \otimes \boldsymbol{u} \in \mathcal{U}$.

(zero vector) $\hat{0} \in S$ because $\hat{0} = \hat{0}^T$

(close under $\oplus$) $\hat{A}, \hat{B} \in S \Rightarrow \hat{A} = \hat{A}^T, \hat{B} = \hat{B}^T$

$\Rightarrow (\hat{A} \oplus \hat{B})^T = \hat{A}^T \oplus \hat{B}^T = \hat{A} \oplus \hat{B}$

true because

$\Rightarrow \hat{A} + \hat{B} \in S$

$\hat{A} = \hat{A}^T, \hat{B} = \hat{B}^T$

(closed under $\otimes$) $\lambda \in \mathbb{F}, \hat{A} \in S \Rightarrow \hat{A}^T = \hat{A}$

$(\lambda \otimes \hat{A})^T = \lambda \otimes \hat{A}^T = \lambda \otimes \hat{A}$

$\therefore \lambda \otimes \hat{A} \in S$

Tutorial we look @ conjugate transpose